

Computer Science & Information Technology

Discrete Mathematics

DPP: 1

Set Theory & Algebra

Q1 How many of the following statements are true?

- I. $\phi \in \phi$
- II. $\phi \subset \phi$
- III. $\phi \subseteq \phi$
- IV. $\phi \in \{\phi\}$
- V. $\phi \subset \{\phi\}$
- VI. $\phi \subseteq \{\phi\}$

Q2 If a set A has 63 proper subsets, then what is the cardinality of A?**Q3** If a set A has 64 subsets of odd cardinality, then what is $|A|$?

- (A) 6
- (B) 63
- (C) 7
- (D) 128

Q4 How many subset of $\{1, 2, 3, \dots, 11\}$ contains at least one even integer?**Q5** Let $A = \{1, 2, 3, 4, 5, 7, 8, 10, 11, 14, 17, 18\}$. How many subsets of A contains exactly six elements?**Q6** Let $A = \{2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15\}$. How many six-elements subsets of A contains exactly four even integers and two odd integers?**Q7** Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$.

How many subsets of A contains only odd integers?

Q8 Suppose A, B, C and D are subsets of U (the universe) with A as a subset of B and C as a subset of D i.e., $A \subseteq B$ and $C \subseteq D$.

Now consider the following statements

- I. $A \cap C \subseteq B \cap D$
- II. $A \cup C \subseteq B \cup D$

Which of the following option is correct?

- (A) Only I is true
- (B) Only II is true
- (C) Neither I nor II is true
- (D) Both I and II are true

Q9 Let $A = \{1, 2, 3, \dots, 15\}$. How many subsets of A contains all of the odd integers in A?**Q10** Let A and B be sets where $A = \{x \mid x^2 - 7x = -12\}$ and $B = \{x \mid x^2 - x = 6\}$. Which of the following is true regarding $A \cup B$ and $A \cap B$?

- (A) $A \cup B = \{5\}$ and $A \cap B = \{-2, 3, 4\}$
- (B) $A \cup B = \{3\}$ and $A \cap B = \{-2, 3, 4\}$
- (C) $A \cup B = \{-2, 3, 4\}$ and $A \cap B = \{3\}$
- (D) $A \cup B = \{2, 3, 4\}$ and $A \cap B = \{5\}$

Answer Key

Q1 4~4**Q2** 6~6**Q3** (C)**Q4** 1984~1984**Q5** 924~924**Q6** 225~225**Q7** 63~63**Q8** (D)**Q9** 128~128**Q10** (C)

Hints & Solutions

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Q1 Text Solution:

- I. $\phi \in \phi$ is false because the empty set has no members.
- II. $\phi \subset \phi$ is false because the empty set is not a proper subset of itself.
- III. $\phi \subseteq \phi$ is true because the empty set is a subset of every set and so it is a subset of itself.
- IV. $\phi \in \{\phi\}$ is true because ϕ is a member here.
- V. $\phi \subset \{\phi\}$ is true because the empty set is a proper subset of itself.
- VI. $\phi \subseteq \{\phi\}$ is true because the empty set is a subset of every set.

Q2 Text Solution:

If a set has n elements then the number of subsets will be 2^n and the number of proper subsets will be $2^n - 1$.

A has 63 proper subsets so,

$$2^n - 1 = 63$$

$$\Rightarrow 2^n = 63 + 1$$

$$\Rightarrow 2^n = 64$$

$$\Rightarrow 2^n = 2^6$$

$$\therefore n = 6$$

The cardinality of A is 6.

Q3 Text Solution:

The number of subsets for $\{1, 2, 3, \dots, n\}$ with odd cardinality is 2^{n-1} .

Now given that,

$$2^{n-1} = 64$$

$$\Rightarrow 2^{n-1} = 2^6$$

$$\Rightarrow n - 1 = 6$$

$$\Rightarrow n = 7$$

Q4 Text Solution:

There are 2^{11} subsets for the set $\{1, 2, 3, \dots, 11\}$.

Out of which, there are 2^6 subset for the set $\{1, 3, 5, 7, 9, 11\}$ that contains none of the even integers.

Hence, there are $2^{11} - 2^6 = 1984$ subsets that contains at least one even integer.

Q5 Text Solution:

If we choose 6 elements from a set of 12 elements then we can do it in ${}^{12}C_6 = 924$ ways.

Q6 Text Solution:

The set A can be bifurcated into Even and Odd subsets as follows:

Even integers = $\{2, 4, 6, 8, 10, 12\}$

Odd integers = $\{5, 7, 9, 11, 13, 15\}$

Out of 6 even integers, we can choose 4 even integers in 6C_4 ways.

Similarly out of 6 odd integers, we can choose 2 odd integers in 6C_2 ways.

Therefore, the total number of required subsets =

$${}^6C_4 * {}^6C_2 = \frac{6^3 \times 5 \times 4!}{2 \times 4!} \times \frac{6^3 \times 5 \times 4!}{4! \times 2} = 15 * 15 = 225$$

Q7 Text Solution:

The set of odd integers: $\{1, 3, 5, 7, 9, 11\}$

Total subsets (including the empty set) = $2^6 = 64$

Subtracting the empty set: $64 - 1 = 63$

Thus, the number of subsets containing only odd integers is **63**.

Q8 Text Solution:

Statement I: $A \cap C \subseteq B \cap D$

- Since $A \subseteq B$, every element in $A \cap C$ is also in $B \cap C$.
- Since $C \subseteq D$, every element in $B \cap C$ is also in $B \cap D$.
- Thus, $A \cap C \subseteq B \cap D$ is **true**.

Statement II: $A \cup C \subseteq B \cup D$

- Since $A \subseteq B$, every element in A is also in B, so $A \cup C \subseteq B \cup C$.
- Since $C \subseteq D$, every element in C is also in D, so $B \cup C \subseteq B \cup D$.
- Thus, $A \cup C \subseteq B \cup D$ is **true**.

Q9 Text Solution:

In the given set $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$, there are 8 odd integers and 7 even integers.

For all the even integers we have two choices whether to include it or not.

Therefore, total possibilities = $2^7 = 128$

Q10 Text Solution:

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$$\begin{aligned}x^2 - 7x &= -12 \Rightarrow x^2 - 7x + 12 = 0 \Rightarrow (x-4)(x-3) = 0 \Rightarrow x = 4, x = 3. \\x^2 - x &= 6 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x-3)(x+2) = 0 \Rightarrow x = 3, x = -2.\end{aligned}$$

Consequently, $A \cap B = \{3\}$
and $A \cup B = \{-2, 3, 4\}$.

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